PT MERDEKA TSINGSHAN INDONESIA

STANDARD OPERATING PROCEDURE

RING COOLER POST 环冷机岗位

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1. Purpose 目的

This procedure covers the operator's responsibility for visual inspection, start up and shutdown of the ring cooler post under complete closure.

此程序涵盖操作员对完全关闭下的环冷机岗位进行目视操作前检查、启动运行和停车的责任

2. Functional Description 功能说明

It is the responsibility of the operator to ensure that the ring cooler post is visually inspected before and after each shift to determine whether it can start and operate safely, or whether other activities (such as maintenance and repair or other operator tasks) must be performed before starting, to determine the sequential operation method of the ring cooler post and smooth operation, Stop order of normal stops and emergency stops.

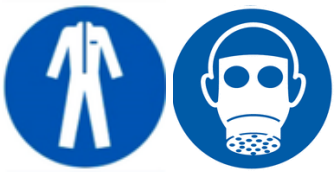
操作员有责任确保在从停机启动之前和每个班次开始时对环冷机岗位进行目视检查，确定是否可以启动并安全运行，或者是否必须在开始启动之前执行其他活动(例如维护修理或其他操作员任务)，确定环冷机岗位的按照顺序启动的操作方法及平稳运行，正常停车和紧急停车的停止顺序。

3. References 参考

- MTI-CP-SOP-Post Technical Operation Procedures.MTI-CP-SOP岗位操作规程
- Feed Hopper Partition Fan(4305-BLO-004/005). 给料斗隔墙冷却风机 (4305-BLO-004/005)
- Pellet Cooling Fan(4305-BLO-001/002/003). 球团冷却风机 (4305-BLO-001/002/003)
- Ring Cooler(4305-COO-001). 环冷机 (4305-COO-001)
- High temperature pulse single machine bag filter(4305-DUC-201). 高温脉冲单机除尘器 (4305-DUC-201)
- High temperature resistant belt conveyor(4305-BCV-001). 链斗输送机 (4305-BCV-001)

4. PPE 个人保护装备





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5. Hazards & Control 危害与控制

Description of Hazard / Environmental Issue危害	Control Strategy控制
1. Mechanical injury; 2. Dust; 3. Electric shock; 4. Slippery conditions and a trip to the danger; 5. Asphyxia; 6. Empyrosis; 7. Object falling from high altitudes; 8. High temperature; 9. Poisoning 10. Noise. 1. 机械伤害； 2. 灰尘； 3. 触电； 4. 湿滑的条件，绊倒危险； 5. 窒息； 6. 烫伤； 7. 高空坠物； 8. 高温； 9. 中毒； 10. 噪音。	1. Install a protective cover on the rotating part of the equipment; 2. Wear dust mask as required; 3. Do not contact the electrical switch with wet hands during operation, Regular inspection of the equipment leakage protection device is complete, effective; 4. Please hold the handrail up and down the stairs; 5. Clear clearance or maintenance operations; 6. Wear good labor protection goods; 7. Wear labor protection goods and check the safety of stacking items on the operation platform; 8. Leave the high temperature environment and rest in a cool and ventilated place; 9. Wear antigas masks as required; 10. Wear earplugs to protect your hearing. 1. 设备的转动部位安装防护罩； 2. 按要求佩戴防尘口罩； 3. 操作时不要湿手接触电器开关，定期检查设备漏电保护装置齐全、有效； 4. 上下楼梯请扶好扶手； 5. 清仓或维修作业执行审批程序； 6. 穿戴好劳保护品； 7. 穿戴好劳保护品，检查各平台的物品堆放安全性； 8. 脱离高温环境，在阴凉通风处休息； 9. 按要求佩戴防毒口罩； 10. 佩戴好耳塞，保护听力。

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6. Equipment 工具/ 设备

4305-MOT-001	Main Motor 环冷机主电机
4305-MOT-002	Discharge Valve Motor 环冷机卸矿阀电机
4305-MOT-003	Accident Motor 环冷机事故电机
4305-ACI-001 ~ 009	Double Layer Ash Valve 双层卸料阀
	Clean the site tools 清扫现场工具

7. Procedure 程序

№	Stage	Key Points
1	General Check 普查	1. Confirm the pre-job technical training. 2. Wear suitable labor protection products, check, and confirm that there are no unsafe state or safety risks on the site. 3. Check and clean the equipment and the site sanitation. 4. Contact with the process before and after and give clear instructions. 1. 确认经过岗前技术培训; 2. 穿戴好适合的劳保护品，检查确认现场没有存在不安全状态或安全隐患; 3. 检查并打扫设备、现场卫生; 4. 与前后工序取得联系，并发出明确指令。
2	Inspect Feed Hopper Partition Fan(4305-BLO-004/005), Pellet Cooling Fan(4305-BLO-001/002/003), High Temperature Pulse Single Machine Bag Filter(4305-DUC-201) Bring Your Own Fan 检查给料斗隔墙冷却风机(4305-BLO-004/005),球团冷却风机(4305-BLO-001/002/003),高温脉冲单机除尘器(4305-DUC-201)自带风机	1. Manual turning 2~3 turn, flexible rotation, no scraping, stuck and other abnormal phenomena. 2. The fan air outlet valve is normal, and the fan inlet valve is closed. 3. Whether the coupling concentricity meets the requirements. 4. Check whether the lubricating oil level of the fan bearing box is normal. 5. Close the power switch, first click (press the operation button and quickly press the stop button) to confirm that the steering is correct, no noise, no vibration, and other abnormal conditions. 6. After the fan is started and running, observe the change of the fan sound, check the temperature and noise of the fan bearing and the motor bearing, and check the fan vibration (including the impeller). 1. 手动盘车2 ~ 3转，转动灵活，不得有刮蹭、卡住等异常现象;

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№	Stage	Key Points
		2. 风机出风阀门正常，风机进口阀门关闭； 3. 联轴器同心度是否符合要求； 4. 检查风机轴承箱润滑油位是否正常； 5. 合上电源开关，先点动(按运转按钮后快速按停止按钮)确认转向正确、无杂音、无振动等异常情况； 6. 风机启动运行后，观察风机声音的变化，检查风机轴承、电机轴承的温度和噪声，检查风机振动(包括叶轮)。
3	Inspect Ring Cooler(4305-COO-001) 检查环冷机 (4305-COO-001)	1. Check whether the motor, reducer, active friction wheel and passive friction wheel are normal. 2. Check whether the castor bar of each car is damaged, whether the net screen is broken, whether there are cracks in the side plate, whether the refractory material falls off, and whether there are gaps in the inner and outer rings. 3. Ensure that there are no personnel and any obstacles around the equipment and in the operating parts. 4. Check whether the action of the double-layer ash discharge valve is flexible and whether there is a foreign body in the bellows to block the discharge outlet. 5. Check whether the discharge outlet is blocked by foreign bodies. 6. Check the use of refractory materials of the trolley rail plate, smoke cover and partition wall, whether there is falling off and damage. 7. In the wind and water supply state, check whether the wind, water cooling partition wall and water-cooling beam devices are working normally. 8. Check whether the population is tightly closed tightly. 9. Check whether the centralized lubrication system is normal. 10. Check whether the running track of the roll arm and winding rail in the unloading area is normal. 11. Check whether the rotation of the trolley support wheel is flexible. 12. Check whether the side gear wheel rotation of the trolley is flexible. 13. Check whether the filling sand sealing between the trolley block plate and the cover is good.

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№	Stage	Key Points
		1. 检查电机、减速机、主动摩擦齿轮、被动摩擦齿轮是否正常； 2. 检查各台车的篦条有无破坏、篦板有无破裂、台车侧板有无裂隙、耐火材料有无脱落，内、外环有无缝隙； 3. 确认设备周围及运转部位没有非本岗位人员和任何障碍物； 4. 检查双层卸灰阀动作是否灵活及风箱内是否有异物堵塞排料口； 5. 检查出料口是否有异物堵塞； 6. 检查台车栏板、烟罩和隔墙的耐火材料使用状况，是否有脱落和损坏； 7. 在供风、水状态下，检查各风、水冷隔墙和水冷梁装置是否工作正常； 8. 检查观察人口关闭是否严实； 9. 检查集中润滑系统是否正常； 10. 检查卸料区域辊臂与曲轨运行轨迹是否正常； 11. 检查台车支撑托轮转动是否灵活； 12. 检查台车侧挡轮转动是否灵活； 13. 检查台车拦板与罩子间的充填砂密封是否良好。
4	Inspect High temperature pulse single machine bag filter(4305-DUC-201) 检查高温脉冲单机除尘器(4305-DUC-201)	1. Whether the electrical high and low voltage system and PLC control are normal. 2. Whether the inlet valve of the filter chamber of each unit is open. 3. Whether all the solenoid pulse valves are normal, and the action sequence is correct. 4. All measurement circuits are normal and signal. 5. All control valves are in the automatic mode and in the desired position. 6. Open all inspection entrances and doors to check whether the equipment moving parts are stuck in dust or external objects; after inspection, close all inspection population and doors and check all pipes and equipment inspection holes for closure. 7. Whether the sealing strip of the air chamber is sealed, and whether the fastening bolt of air box sealing cover is in good condition. 8. Connect the pressure measuring device to the

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№	Stage	Key Points
		pressure measuring port of the bag filter. 9. Check whether the bag back blowing pressure is 0.4MPa~0.5MPa. 10. Start the dust fan and observe the empty load operation resistance. 11. Check whether the water drain trap is normal. 12. Check the operation condition of the ash removal device of the bag filter. 1. 电气高低压系统、PLC控制是否正常； 2. 除尘器各单元过滤室的进口阀门是否都打开； 3. 所有电磁脉冲阀是否正常，动作顺序正确； 4. 所有测量回路是否正常、有信号； 5. 所有提升阀处于自动模式和需要的位置； 6. 打开所有检查入口、门，检查设备运动部件是否被灰尘或外来物卡住；检查后，关闭所有检查人口、门，并检查所有管道和设备检查孔是否关闭； 7. 反吹风气室密封条是否是密封完好，气箱密封盖紧固螺栓是否完好； 8. 接通测压装置到布袋除尘器的测压口； 9. 检查布袋反吹风压力是否是0.4MPa～0.5MPa； 10. 启动除尘器自带风机，观察空负荷运行阻力； 11. 检查疏水阀是否正常； 12. 检查布袋除尘器排灰装置工作情况。
5	Inspect High temperature resistant belt conveyor(4305-BCV-001) 检查链斗输送机(4305-BCV-001)	1. Check whether the wear and material between the head and tail wheels. 2. Check whether the connection is reliable and whether the bolts are fastened. 3. Check that the lubricating oil level of the reducer is appropriate, and the lubrication and sealing of all bearings are good. 4. Check whether the roller, chain shaft and so on are easy and flexible. 5. Check whether the electrical facilities and control system are in good condition, and whether the running direction of the motor is correct. 6. Check whether the displacement amount of the tension chain axis meets the requirements. 7. Check whether the tensioning device of the conveying chain is in good condition, whether the

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№	Stage	Key Points
		tensioning condition is within a reasonable range, and whether the conveying chain is deviated phenomenon. 8. Check the wear condition of the transport chain and the track. 9. Check whether the engagement of the sprocket and the chain link is intact. 10. Check whether the safety protection device is complete and reliable, and whether the emergency stop device, deviation correction and speed measurement sensors are intact. 1. 检查首、尾托轮磨损情况与链斗间是否夹有物料； 2. 检查各处连接是否可靠，螺栓是否紧固； 3. 检查减速机润滑油油位适当，各部轴承润滑与封闭良好； 4. 检查各托辊、链轴等是否轻松、灵活； 5. 检查电器设施和控制系统是否完好，电机的运转方向是否正确； 6. 检查张紧链轴的位移量是否合乎要求； 7. 检查输送链的张紧装置是否完好，张紧情况是否在合理的范围内，输送链有无跑偏现象； 8. 检查输送链、轨道磨损情况； 9. 检查链轮与链节的啮合情况是否完好； 10. 检查安全防护装置齐全、可靠，急停装置、纠偏和测速传感器是否控制灵敏。
6	Check the other Equipment and Facilities 检查其它设备设施	1. Chute (4305-CHE-001) a. Check whether the feeding chute is in good condition. b. Check whether the blocking material detection switch of the lower feeding chute is normal. 1. 溜槽(4305-CHE-001) a. 检查下料溜槽是否完好； b. 检查下料溜槽的堵料检测开关是否正常。 2. Star Discharger a. Check whether the rotation direction of the star valve rotor at the ash outlet of the P (4305-DUC-201) is correct.

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№	Stage	Key Points
		2. 星型卸料阀 a. 检查高温脉冲单机除尘器(4305-DUC-201)排灰口的星型阀转子的旋转方向是否正确。 3. Motor a. Check whether the main motor (4305-MOT-001), discharge valve motor (4305-MOT-002) and accident motor (4305-MOT-003) are grounded intact, and the rotation direction is correct. 3. 电机 a. 检查环冷机主电机(4305-MOT-001)、环冷机卸矿阀电机(4305-MOT-002)和环冷机事故电机(4305-MOT-003)接地是否完好，旋转方向是否正确。 4. Valve a. Check feed hopper partition fan (4305-BLO-004/005) outlet butterfly valve (400BF),pellet cooling fan (4305-BLO-001/002/003) inlet gate valve (40GA) exit butterfly valve (130BF),Whether high temperature pulse single machine bag filter inlet valve is normal, Whether the cooling water valve (150GA, 80GA, 100GA, 150GA) is normal. 4. 阀门 a. 检查给料斗隔墙冷却风机(4305-BLO-004/005)出口蝶阀(400BF)、球团冷却风机(4305-BLO-001/002/003)进口闸阀(40GA)出口蝶阀(130BF)，除尘器进口阀是否正常，冷却水闸阀(150GA，80GA，100GA，150GA)是否正常。 5. Flue-Dust Retainer a. Whether the ash collector of the ring cooler (4305-COO-001) is smooth. 5. 集灰斗 a. 环冷机(4305-COO-001)集灰斗放汇是否顺畅。

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№	Stage	Key Points
		6. Double Layer Ash Valve (4305-ACI-001 ~ 009) a. Turn on the power supply and check whether the motor steering direction is consistent with the arrow mark. b. Check whether the electric double-layer ash discharge valve weight block stops and returns, whether the swing rod is loose and falls off, and whether the flange connection bolt is loose and falls off. c. Fill the bearing base with lubricating grease (general lithium grease, 30g /) d. Check whether the cycloid needle wheel reducer is normal, and if the lubricating oil level is normal. 6. 双层卸料阀(4305-ACI-001 ~ 009) a. 接通电源，检查电机转向与箭头标示方向是否一致； b. 检查电动双层卸灰阀配重块止退销、摆杆是否松动脱落，法兰连接螺栓是否有松动脱落； c. 轴承座加注润滑油脂(通用锂基脂,30g/处)； d. 检查摆线针轮减速机是否正常，润滑油位是否正常。
7	Inspect Materials Level 检查物料料位	1. Check whether the room in the ring cooler (4305-COO-001) has been stored. 2. Check whether the dust bucket of the high temperature pulse single machine bag filter (4305-DUC-201) is dust. 3. Check the stock in the chain conveyor (4305-BCV-001) 4. Ring cooler (4305-COO-001) ash bucket whether ash 5. Check whether the cooling circulating water pressure of the water-cooled partition wall and water-cooled beam device is normal. 1. 检查环冷机(4305-COO-001)内个室是否存料； 2. 检查高温脉冲单机除尘器(4305-DUC-201)集灰斗是否存尘； 3. 检查链斗输送机(4305-BCV-001)的链斗里是否存

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№	Stage	Key Points
		料； 4. 环冷机(4305-COO-001)集灰斗是否存灰； 5. 检查水冷隔墙和水冷梁装置冷却循环水压是否正常。
8	Start-Up Of the Ring Cooler Post 环冷机岗位开机	1. No-load start should be guaranteed when starting. 2. The driving sequence is along the material direction, from back to front that is the reverse drive, each equipment needs to be switched on later; At the same time, Confirm that the process behind the ring cooler post has been started and has the feeding conditions: Open the centralized lubricating oil supply system ⟹ Open the cooling water brake valve (150GA, 80GA, 100GA, 150GA), and pass the hopper water cooling partition wall and water cooling beam into the cooling water ⟹ Open the fan butterfly valve (400BF), and start the feper partition wall cooling fan (4305-BLO-004 / 005) ⟹ Open the valve (***) for the bearing through the cooling water of the pellet cooling fan (4305-BLO-001 / 002 / 003), enter the cooling water, open the fan butterfly valve (130BF), and start the pellet cooling fan (4305-BLO-001 / 002 / 003) ⟹ Close the gate valve (***) and the double discharge valve of all bellows (4305-ACI-001~009) ⟹ Open the inlet valve of the dust collector (***), open the dust collector induced draft fan, and start the high temperature pulse single machine dust collector (4305-DUC-201) ⟹ Start the chain bucket conveyor (4305-BCV-001) ⟹ Open the main motor (4305-COO-001) (4305-MOT-001) and gradually accelerate to the normal working speed. 1. 启动时应保证空载启动； 2. 开车顺序为沿物料走向，从后往前开即逆向开车，各设备都需要延时开启；同时，确认环冷机岗位后面的工序已经开车并具备给料条件： 开启集中供给润滑油系统⟹打开冷却水闸阀(150GA,80GA,100GA,150GA)，受料斗水冷隔墙和水冷梁通入冷却水⟹开启风机

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№	Stage	Key Points
		蝶阀(400BF)，启动给料斗隔墙冷却风机(4305-BLO-004/005)⟹打开给球团冷却风机(4305-BLO-001/002/003)轴承通冷却水的阀门(***)，通入冷却水，开启风机蝶阀(130BF)，启动球团冷却风机(4305-BLO-001/002/003)⟹关闭卸料斗的闸阀(***)和所有风箱的双层卸料阀(4305-ACI-001～009)⟹打开除尘器进口阀(***)，开启除尘器引风机，启动高温脉冲单机除尘器(4305-DUC-201)⟹启动链斗输送机(4305-BCV-001)⟹开启环冷机(4305-COO-001)主电机(4305-MOT-001)，逐步加速到正常工作转速。
9	The Ring Cooler Post Stops 环冷机岗位停机	1. Orderly shutdown a. The materials in the standby machine should be discharged before stopping. b. Stop in reverse order of driving order, Each equipment needs to stop: Stop ring cooler (4305-COO-001) main motor (4305-MOT-001)⟹Stop chain hopper conveyor (4305-BCV-001)⟹Stop the high-temperature pulse single-machine dust collector (4305-DUC-201), stop the dust collector induced draft fan, and close the inlet valve of the dust collector (***)⟹Open the gate valve (***) of the dump and the double discharge valve of all bellows (4305-ACI-001~009)⟹Stop pellet cooling fan (4305-BLO-001/002/003), close the fan butterfly valve (130BF), close the valve with cooling water for pellet cooling fan (4305-BLO-001/ 002/003) (***)⟹Stop the cooling fan of the feed per partition (4305-BLO-004/005) and close the fan butterfly valve (400BF)⟹ Close the cooling water brake valve (150GA, 80GA, 100GA, 150GA), stop the hopper water cooling partition wall and water cooling beam into the cooling water⟹ Turn off the centralized supply of lubricating oil system⟹ Clean up the site sanitation.

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№	Stage	Key Points
		1. 正常停车 a. 停车时应待机内物料排净后再停车； b. 按照开车顺序逆序顺序停车，各设备都需要延时停止： 停止环冷机(4305-COO-001)主电机(4305-MOT-001)⟹停止链斗输送机(4305-BCV-001)⟹停止高温脉冲单机除尘器(4305-DUC-201)，停止除尘器引风机，关闭除尘器进口阀(***)⟹打开卸料斗的闸阀(***)和所有风箱的双层卸料阀(4305-ACI-001～009)⟹停止球团冷却风机(4305-BLO-001/002/003)，关闭风机蝶阀(130BF)，关闭给球团冷却风机(4305-BLO-001/002/003)轴承通冷却水的阀门(***)⟹停止给料斗隔墙冷却风机(4305-BLO-004/005)，关闭风机蝶阀(400BF)⟹关闭冷却水闸阀(150GA,80GA,100GA,150GA)，停止受料斗水冷隔墙和水冷梁通入冷却水⟹关闭集中供给润滑油系统⟹打扫现场卫生。 2. Emergency Stop a. When an emergency occurs, Close high temperature pulse single machine bag filter (4305-DUC-201), high temperature resistant belt conveyor (4305-BCV-001), discharge valve motor (4305-MOT-002), main motor (4305-MOT-001), pellet cooling fan (4305-BLO-001/002/003), feed hopper partition fan (4305-BLO-004/005), After all stop the operation, Check whether the start / stop button of each device is in the stop state, Turn off the centralized lubricant supply system, Close cooling water brake valve (150GA, 80GA, 100GA, 150GA), feed hopper partition fan (4305-BLO-004/005) fan butterfly valve (400BF), pellet cooling fan (4305-BLO-001/002/003), fan butterfly valve (130BF), hopper gate valve (***) and all bellows double

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№	Stage	Key Points
		discharge valve (4305-ACI-001～009), high temperature pulse single machine bag filter (4305-DUC-201) inlet valve (***), at the same time, the upper level of equipment all need to stop; b. After the accident, the equipment is restored to normal. If production still needs to be conducted, Regardless of the length of time, start the accident motor (4305-MOT-003) for 10 seconds, start the main motor (4305-MOT-001) with accident motor (4305-MOT-003) for 10 seconds; open the cooling water brake valve (150GA, 80GA, 100GA, 150GA), the hopper water cooling partition wall and water cooling beam into the cooling water; the operation shall be restarted in the driving sequence; c. After the accident treatment, the equipment returns to normal. If the production is not necessary, operate according to the normal parking order. 2. 紧急停车 a. 当出现紧急事故，关闭高温脉冲单机除尘器(4305-DUC-201)、链斗输送机(4305-BCV-001)、环冷机卸矿阀电机(4305-MOT-002)、环冷机主电机(4305-MOT-001)、球团冷却风机(4305-BLO-001/002/003)和给料斗隔墙冷却风机(4305-BLO-004/005)，全部停止运行后，检查各设备的启动/停止按钮是否处于停止状态，关闭集中供给润滑油系统，关闭冷却水闸阀(150GA,80GA,100GA,150GA)、给料斗隔墙冷却风机(4305-BLO-004/005)风机蝶阀(400BF)、球团冷却风机(4305-BLO-001/002/003)风机蝶阀(130BF)、卸料斗的闸阀(***)和所有风箱的双层卸料阀(4305-ACI-001～009)、除尘器(4305-DUC-201)进口阀(***)，同时，上一级设备均需停车； b. 事故处理后，设备恢复正常，如仍需要进行生产，无论时间长短再启动时，必须先

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№	Stage	Key Points
		启动事故电机(4305-MOT-003)，运转10秒后，随即以事故电机(4305-MOT-003)启动主电机(4305-MOT-001)逐步加速到正常工作转速；同时，打开冷却水闸阀(150GA,80GA,100GA,150GA)，受料斗水冷隔墙和水冷梁通入冷却水；按照开车顺序重新启动操作； c. 事故处理后，设备恢复正常，如不需要进行生产，按照正常停车顺序进行操作。
10	Permits & Isolation Sheets 许可证和隔离表	1. Only authorised isolators can sign off and cancel permits and registration sheets. 1. 只有经过授权的隔离者才能签署和取消许可证和登记表。

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8. Ring Cooler Post 环冷机岗位 Questionnaire

Declaration of Understanding

I (_____Learner_____) have completed the theory assessment and had all my valid, reliable, and consistent with my level of knowledge in the assessment task.

Signature
Date:

I (_____Assessor_____) have witnessed the above learner complete the theory assessment and believe the assessment has provided a valid, reliable, and consistent result.

Signature
Date:

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Appendix

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Standard Operating Procedure
标准操作程序

Ring cooler posts in the chloride plant
CP 环冷机岗位

SOP Details 程序明细

Processing Department 氯化厂	Procedure Level 程序级别: **3**	Revision Number 修订编号: 0
Date Approved 审批日期:	Date Issued 颁布日期:	Next Review Date 下次审核日期:
Details of Revision 修订明细:		

Revision Details 修订明细

Rev. No 版本号	Date 日期	Section 章节	Page 页码	Revision Description 修订说明	Reviewed By 审核人
1					
2					
3					

Approvals 批准

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Processing Senior Manager 氯化高级经理	Li Jianjun 李建军	Date: 日期: 2024.04.24	Signature 签字	
Safety Department Manager 安全部经理	Airlangga Putra R.J	Date: 日期: 24,04,24	Signature 签字	
Deputy General Manager 副总经理	Li Xinhua 李新华	Date: 日期 2024.4.24	Signature 签字	
General Manager 总经理	Clark Wu 吴迎林	Date: 日期: 2024.4.24	Signature 签字	